Computational Physics: Introductory Course

Assignment 8:

Eigen-Value Problems

Per Jönsson & Jörgen Ekman
Faculty of Technology and Society
Malmö University

What the report should contain

A report should contain:

- The py-files, with carefully commented Python code, that were used to solve the exercises. All code should start with a sentence or two describing what the code is doing. Each py-file should be embedded in the LaTeX or Word document.
- When applicable, you should include the output or plots produced by your code as part of the exercise.
- When you, as part of an exercise, are asked to draw physical conclusions, it is important that these conclusions are included in the report. If you are unable to draw any conclusion, you should explicitly write 'I was not able to draw any conclusion'.

Important things to keep in mind for producing a good report that adheres to the general rules in scientific writing.

- All variables in the report should be in math font, e.g., the distance was $d = 1.5$ (do not write the distance was d = 1.5, wrong).
- All units in the report should be in correct font, e.g., the acceleration is $a = 1.5$ ms^{-2} (do not write the acceleration is $a = 1.5\,ms^{-2}$, wrong).
- Write $a = 1.5 \cdot 10^{-3}$ instead of $a = 1.5 * 10^{-3}$ (wrong, never use * in equations or text)
- If you give computed values, round them to a reasonable number of decimals.
- The font size in plots should be large enough for the axis texts and legends to be read.

Reading instructions for assignment 8

Read sections 11.16, 11.18-11.20, 21.6, 21.7 and 21.9 in the lecture notes

Exercises

1. Look through the commands for handling matrices so that you master them (need not be reported in writing, but is left to you for your own knowledge)
2. Write a py-file that determines and visualizes the eigen oscillations for a string with three mass points as an animation (see code in section 11.20 in the lecture notes). Include a few 'snapshots' in the report.
3. Write a py-file that determines and visualizes the eigen oscillations for a string with 10 mass points as an animation. Also, test to see how the vibrations look like when you take some linear combinations of the eigen oscillations. Again, include a few 'snapshots' in the report.
4. Download `quantumwell.py` (code in section 21.7 available from Canvas) and test the code for solving the wave equation for a particle in a potential well.

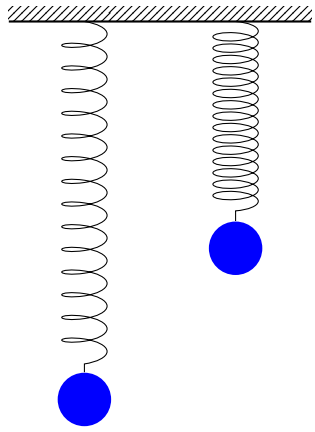


Figure 1: *A particle with mass $m = 1$ attached to a spring will oscillate up and down around its equilibrium position.*

5. A particle with mass $m = 1$ attached to a spring with spring constant k make up a harmonic oscillator, see figure 1.

   Let x be the position of the particle relative to the equilibrium position. In classical mechanics, the equation of motion is

   $$\frac{d^2x}{dt^2} = -kx,$$

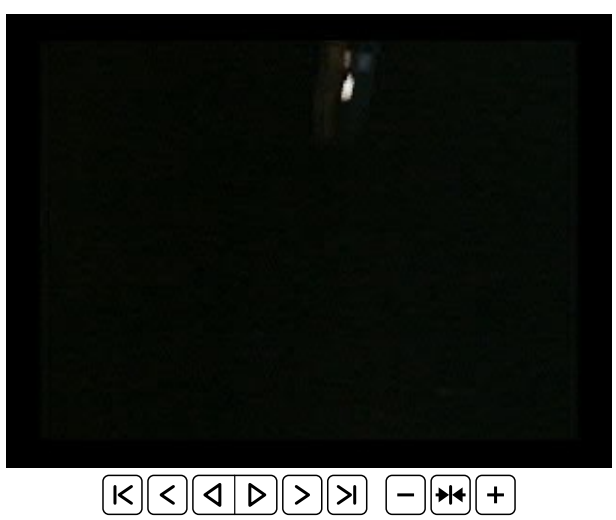
   where k is the spring constant. Solving the equation, we see that the position is given by a sinusoidal function with angular frequency $\omega = \sqrt{k}$.

   We can also see the oscillations back and forth as a particle moving in a parabolic potential

   $$V(x) = \frac{1}{2}kx^2,$$

   where the connection between the two views is given by the potential-force relation $F = -\frac{dV}{dx}$.

   Below we have a short video of a particle attached to a spring oscillating around its equilibrium.



From the video it is clear that the highest probability to find the particle is close to its turning points where the speed is low. The lowest probability to find the particle is around its equilibrium position, where the potential energy is zero and the speed is at its highest. We bear this in mind as we will compare it with the quantum mechanical interpretation.

(a) Modify `quantumwell.py` and solve the Schrödinger equation for the potential above (harmonic oscillator) with $k = 1$ and determine the energies (eigenvalues) and wave functions (eigenvectors) for the 10 lowest states. Take the solution interval large enough to ensure that the wave functions for all the states have fallen off to zero. Plot the wave functions and the probability distributions. When you plot the wave functions and the probability distribution, you should *not* plot the potential in the same plot. For this reason, you should comment out the following two lines in the modified `quantumwell.py` program

```
ax.plot(x,1 + 2*u/(np.max(u)-np.min(u)),"b")
```

and

```
ax.set_xlim([-3*x_half,3*x_half])
```

(b) Compare the computed energies for the harmonic oscillator with the analytical ones

$$E_n = 0.5 + n, \qquad n = 0, 1, 2, \ldots$$

Increase the number of grid points to say, 5000, instead of 500 and see how the accuracy of the computed energies improve.

(c) Compare the probability distributions in classical mechanics and quantum mechanics. In classical mechanics, the highest probability to find the particle is close to the turning points. Explain how is this reflected in the quantum mechanical probability distribution. To be able to explain, study the probability distribution also for some of the higher states.

To end this exercise, we want to make a comment about a major difference between classical mechanics and quantum mechanics. In classical mechanics, the total energy for a particle that is at rest at its equilibrium point (at rest in the bottom of the potential) is zero. In quantum mechanics, the lowest possible energy is not zero but $E_0 = 0.5$ atomic units. The particle can never be at rest at the bottom of the potential due to what is called Heisenberg's uncertainty relation that states that a precise determination of the position (we have made a measurement and know that the particle is at the equilibrium position or at the bottom of the potential) by necessity implies an uncertainty of the momentum p and the particle will be associated with a positive kinetic energy. Turning things around, if we have made a precise measurement of the momentum p (kinetic energy) and know that this is zero, then the position will be uncertain, and the particle may be somewhere away from the equilibrium point with a positive potential energy. The particle can never have zero energy! Amazing!

Physics background

Concepts of classical physics – particle

A particle is a discrete entity with definite position and momentum. It moves in accordance with Newton's laws of motion. As an example, we take the motion of a planet bound to the sun by the gravitational force ($1/r^2$). By applying Newton's law of motion and solving the resulting ordinary differential equation, we found, in assignment 5, that the planet moves in an elliptical trajectory, with the sun in one of the focuses.

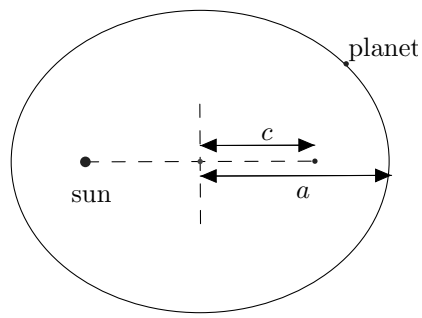


Figure 2: *The orbit of a planet is an ellipse, where one focus of the ellipse is the sun. The eccentricity is equal to the distance between a focus and the center c of the ellipses divided by the semimajor axis a. That is,* $e = c/a$.

During the motion, both the total energy

$$E_{\text{tot}} = E_{\text{kin}} + E_{\text{pot}}$$

and the angular momentum

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}$$

are preserved (constants of motion).

Concepts of classical physics – wave

Waves are also a classical concept. A wave is an extended quantity present at every point in space. There are several types of waves such as mechanical waves propagating in materials, sound waves consisting of pressure variations in air, and electromagnetic waves consisting of electric and magnetic fields that satisfy Maxwell's equations. Waves are associated with diffraction and interference effects `https://en.wikipedia.org/wiki/Diffraction`. We can also have a formation of standing waves that give rise to discrete energies, compare section 11.20 in the lecture notes.

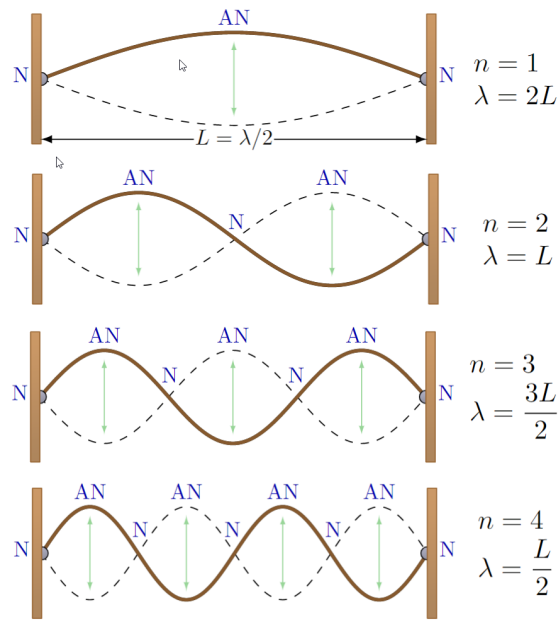


Figure 3: *Standing waves in classical physics give rise to discrete energies.*

Limitations of classical physics

In the beginning of the 20:th century, it was clear that classical physics could not explain a number of phenomena and processes. Most importantly, it could not explain the properties of the atoms. The view of an atom as a 'planetary system in miniature' governed by classical mechanics (Newton's laws) fails, not least due to the fact that the atomic energy levels can take on only well-defined discrete energies characteristic for the specific element. This is in sharp contrast to classical mechanics, where the energy can be in a continuous range depending on the initial conditions.

Quantum mechanics – combining particles and waves

The limitations of classical physics can be overcome if one accepts the fact that a system can have both particle and wave properties. In 1905 Einstein suggests that light with frequency f and wavelength λ should be seen as a stream of particles (photons) with energy

$$\epsilon = hf = \frac{hc}{\lambda} \quad \text{(we use } c = f\lambda\text{)}$$

and momentum

$$p = \frac{\epsilon}{c} = \frac{h}{\lambda}$$

where $h = 6.626 \times 10^{-34}$ Js is Planck's constant. The photon concept explains, among other things, the photoelectric effect `https://en.wikipedia.org/wiki/Photoelectric_effect` and the energy density observed from black-body radiation `https://en.wikipedia.org/wiki/Black-body_radiation`. If light is a stream of particles (photons) what about interference as displayed through Young's double slit experiment? It turns out that interference is not a macroscopic property, and experiments shows that each photon interferes only with itself due to the quantum uncertainty of which path through the optical apparatus it takes. What we observe as an interference pattern is the accumulated sum of many single photon interference events, read more at `https://sciencedemonstrations.fas.harvard.edu/`

`files/science-demonstrations/files/single_photon_paper.pdf`

If waves behave like particles (photons) it is reasonable to expect that material particles behave like waves. In 1923 Louis de Broglie suggested that a material particle with momentum p can act as a wave with wavelength

$$\lambda = \frac{h}{p}.$$

The wavelength is called the de Broglie wavelength. The relativistic energy ϵ for a particle is given by

$$\epsilon^2 = m^2c^4 + p^2c^2.$$

This gives

$$p^2c^2 = \epsilon^2 - m^2c^4 \Leftrightarrow pc = \sqrt{(\epsilon - mc^2)(\epsilon + mc^2)}$$

and thus

$$\lambda = \frac{hc}{\sqrt{(\epsilon - mc^2)(\epsilon + mc^2)}}.$$

For an ultra relativistic particle (kinetic energy much larger than the mass energy mc^2)

$$\lambda = \frac{hc}{\sqrt{(\epsilon - mc^2)(\epsilon + mc^2)}} \approx \frac{hc}{\epsilon}.$$

For a non-relativistic particle we have

$$\epsilon = mc^2 + Em$$

with $E = p^2/2m$ the ordinary kinetic energy. This gives

$$\lambda = \frac{h}{\sqrt{2mE}}.$$

Experiments with particles confirm that they indeed have wave properties in accordance with the relations above, and if one shoots single particles such as electrons through a double slit one gets the same type of interference pattern as for photons `https://en.wikipedia.org/wiki/Double-slit_experiment`, see figure 4.

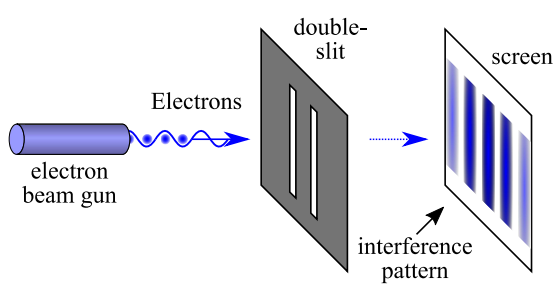


Figure 4: *Photons or particles of matter (like an electron) produce a wave pattern when two slits are used.*

It was soon realized that the wave property of material particles could not fully be described by de Broiglie's theory. If there were waves, then there must, in the same way as for electromagnetic waves, be

an equation governing these waves. In 1926 Erwin Schrödinger formulated such an equation. A single particle moving in a time-independent potential $V(x)$ is described by a so-called wave function $\Psi(x)$ that satisfies

$$-\frac{\hbar^2}{2m}\frac{d^2\Psi(x)}{dx^2} + V(x)\Psi(x) = E\Psi(x),$$

where $\hbar$ is Planck's constant and m the mass of the particle. E is the total energy of the particle, kinetic plus potential energy. The wave function has a probability interpretation and for bounded states, the probability to find the particle in an interval dx around x is given by $|\Psi(x)|^2\,dx$. The probability interpretation gives that the eigenfunction should be normalized

$$\int_{-\infty}^{\infty} |\Psi(x)|^2\,dx = 1,$$

which, in turn, means that the wave function must satisfy $\Psi(-\infty) = \Psi(\infty) = 0$.

The wave equation for a particle has the same properties as the equation for the mass points on a string. The discrete frequencies and vibration modes for the mass points on the string are consequences of the zero amplitude for the endpoints of the string. In the same way, the discrete energies for a particle in a potential are consequences of the wave function being zero at the end points.

The Schrödinger equation generalizes to three dimensions, and applied to hydrogen, it perfectly predicts the discrete energy levels found experimentally, one of the greatest triumphs of the 20:th century physics. Applied to many electron atoms and molecules, the Schrödinger equation is one of the cornerstones in modern physics and chemistry.